Principal Test

Wednesday 12 March 2025 11:35

Principal Engineer take home test:

We would like you to put together a high level architecture document proposing a solution for **one** of the two scenarios presented below:

1. An airport trip planner, that offers recommendations on restaurants and shops I could visit based on my preferences, my flight time and potential sales happening at shops.
    a. Assume we have:
        i. A service that will provide you with my selected preferences and flight time
        ii. A service that provides a list of shops and restaurants in the airport along with some keywords that describe the shop - these keywords are not standardised and provided by the brands themselves
            1) Eg: McDonalds - fast food, cheap, takeaway
            2) Coach - luxury, fashion, service
2. A tool to predict airport lounge occupancy, helping our users choose which is the best lounge (or flight) to book
    a. Assume we have:
        i. A service that provides real time indication of who has entered a lounge but not if they have left
        ii. A service that provides my preferences and flight time
        iii. A service that provides all the flights to and from the airport

Outputs:

- A diagram presenting your back end, front end and infrastructure technology choices.
- A brief description on why each technology was chosen.
- The strengths and trade offs in your design.

Notes:

- We will be reviewing this solution together in the next stage of the interview process
- Focus on pre built ai solutions rather then training your own model
- You are welcome to use any tool to assist you in creating your solution (AI included)